

JOE BOWEN

Chapel Hill, NC — Open to Relocation

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Summary

Statistics and Analytics student with experience in machine learning, data engineering, and operations analytics. Proven ability to turn complex datasets into actionable insights through Python, SQL, and Power BI. Experienced in automation, computer vision, and cross-functional collaboration within manufacturing environments.

Technical Skills

Languages: Python, R, SQL, Java, HTML, CSS, JavaScript, L^AT_EX, MATLAB

Dev Tools: Databricks, Docker, Git, GitLab, VS Code, Anaconda, Google Colab, GPT

Experience

Corning Inc.

Summer 2025

Digital Transformation Intern

Hickory, NC

- Engineered and deployed computer vision models using Ultralytics YOLO on Databricks for automated quality-control inspection, curating datasets and managing version control via GitLab to streamline development workflows. Reduced Beckhoff PLC cycle time by ~20 ms and improved model confidence scores by ~60% through optimization of detection pipelines and validation testing.
- Designed and implemented Ignition (SCADA) and Power BI dashboards integrating 2,500+ machine tags and datasets exceeding 1.6M rows, enabling real-time engineering insights and process monitoring.
- Led Extract, Transform, Visualize project, extracting data from Beckhoff PLCs and thermal imaging systems, transforming via Python in an RPA environment, and delivering production-ready Power BI reports to optimize manufacturing recipe tuning.

Corning Inc.

Summer 2024

Operations Intern

Winston-Salem, NC

- Automated aggregation of 50,000+ SQL data cells into dynamic Power BI dashboards using Power Query, reducing meeting preparation time by up to 50% and forecasting time from 30 to 10 minutes.
- Applied DMAIC methodology to standardize plant-wide material storage and disposal processes, improving operational consistency and compliance.
- Developed automated data workflows using Microsoft Office and Power Automate to support long-term integration of new manufacturing processes into enterprise systems.

CVS Health

Sep 2024 – Mar 2026

Shift Supervisor

Chapel Hill, NC

- Supervise daily operations for a high-volume retail store, overseeing inventory, cash handling, and regulatory compliance while leading team members to maintain operational efficiency and resolve real-time issues under pressure.

Projects

Uncertainty-Aware LLM Training | *Python, PyTorch, GPT-2, Google Colab*

Fall 2025

- Retrained a toy GPT-2 language model on domain-specific medical QA data to evaluate how alternative loss functions affect uncertainty calibration and out-of-domain behavior.
- Implemented and compared standard cross-entropy, Linearly Adaptive Cross-Entropy (LACE), and Generalized Kullback–Leibler (GKL) objectives, averaging results across 3 trials.
- Used Monte Carlo Dropout at inference time to estimate epistemic uncertainty on unseen out-of-domain prompts, showing LACE increased OOD uncertainty relative to standard cross-entropy while GKL led to over-uncertainty.

ASA DataFest Finalist | *R, Python, Data Wrangling, Statistical Analysis*

April 2025

- Collaborated on a four-person team to clean and analyze ~200,000 leasing records, identifying market trends to support tenant decision-making.
- Presented findings to 150+ students and industry professionals, translating complex results into actionable recommendations.

Education

University of Texas at Austin

Expected 2028

M.S. - Computer Science (Online)

Austin, TX

University of North Carolina at Chapel Hill

B.S. - Statistics and Analytics

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